

ANUPAM VISHWAKARMA

AI/ML Engineer | Generative AI | Python Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer and Python Developer with 7+ years of proven experience architecting and delivering enterprise scale Generative AI and machine learning solutions across publishing, energy, cybersecurity, and fintech sectors. Core expertise spans LLM pipeline design, RAG architecture, computer vision, and full-stack AI application development on Google Cloud Platform. M.Tech graduate of IIIT Allahabad.

KEY SKILLS

Languages: Python, Java, C/C++, JavaScript, HTML5/CSS, Google Apps Script
GenAI & LLMs: OpenAI GPT-4, Gemini, Claude, Prompt Engineering, Fine-Tuning
RAG & Vector DBs: LangChain, Chroma DB, Pinecone, FAISS, Semantic Search, Embeddings
ML / DL: Scikit-learn, TensorFlow, PyTorch, YOLO v7, TrOCR, Hugging Face Transformers
NLP: Text Classification, NER, Summarization, spaCy, NLTK, Topic Modeling
Web Frameworks: Django, Django REST Framework, FastAPI, Streamlit
Cloud & DevOps: Google Cloud Platform (GCP), Docker, CI/CD Pipelines
Databases: PostgreSQL, MySQL, MongoDB, Redis, SQL Server
AI Observability: Langfuse, Grafana, Prometheus, Model Monitoring, Cost Optimization
Tools & Agile: JIRA, Confluence, Git/GitHub, Scrum, Kanban, Pytest, SonarQube

PROFESSIONAL EXPERIENCE

Springer Nature Technology & Publishing Solutions, Pune *AI/ML Engineer* | April 2023 – Present | Team: 12 | Python · Django · FastAPI · LangChain · OpenAI · Gemini · GCP

- ▶ Architected and shipped 5 production Gen AI systems Alt Text Generator, Taxonomy Classifier, Ethics Compliance Validator, and Budget Optimizer, serving 500+ academic journals and raising accessibility compliance scores by 60%.
- ▶ Developed RAG workflows with LangChain, Chroma DB, and OpenAI/Gemini, enabling semantic search across 1M+ documents and reducing manual editorial review effort by 35%.
- ▶ Launched Augmented Insights Google Workspace Add-on on Google Marketplace, achieving adoption by 1,000+ editors across 12 countries within 3 months of go live.
- ▶ Wired Langfuse, Grafana, and Prometheus into the observability stack, cutting LLM API spend by 25% through prompt optimization and model switching strategies.
- ▶ Processed 500,000+ EBM records via large scale XML to SAP ETL workflow, automating CRM uploads and eliminating 15 hours/week of manual data entry effort.
- ▶ Benchmarked 5+ open-source vision language models against GPT-4V on academic image tasks, selected cost-optimal model that reduced inference expenditure by 60%.
- ▶ Deployed scalable GCP microservices via Docker and FastAPI, sustaining 99.9% uptime SLA across all production AI systems.

MDP INFRA India Pvt. Ltd (Client: MPMKVCL MP State Power Co.), Bhopal *Team Lead / Python Developer / ML Engineer* | Oct 2021 – March 2023 | Team: 24 | Python · Django · DRF · YOLO v7 · TrOCR · PostgreSQL

- ▶ Led a 24-member cross-functional team to design and deliver QCPortal (qcportal.mpcz.in), an enterprise onboarding solution serving 50,000+ vendors, NABL labs, and government contractors.
- ▶ Designed ML based electricity theft identification system on historical NBS billing data, surfacing abnormal consumption patterns and enabling recovery of an estimated ₹2 Cr+ in annual revenue.
- ▶ Assembled a two-stage computer vision workflow YOLO v7 region localization + TrOCR digit extraction for automated meter reading, attaining 90%+ recognition accuracy across 5,000+ field meters.

CyberInfrastructure Pvt. Ltd, Indore *Software Developer (ML & Python)* | March 2021 – Oct 2021 | Team: 5 | Python · FastAPI · Streamlit · Django · ML

- ▶ Spearheaded development of an ML-powered Risk Maturity Assessment tool FastAPI backend, JWT authentication, and Streamlit dashboards processing structured questionnaires for 200+ enterprise clients and generating automated risk scores with one-click PDF reports.
- ▶ Produced 3 secure, modular Python ML classification modules for a US/Israel defense client, achieving zero security incidents across a 6-month classified delivery cycle while meeting strict code-modularity and documentation standards.

Aspirify Enterprise Pvt. Ltd, New Delhi *Python Developer (AI/ML)* | Aug 2020 – March 2021 | Team: 6 | Django · Celery · Redis · ML · Pandas

- ▶ Constructed a distributed network vulnerability scanner with Django, Celery, and Redis, processing 1,000+ Internetworking Protocol endpoints in parallel and reducing scan cycle time from 48 hours to 8 hours.
- ▶ Embedded ML-based anomaly identification into the scanning workflow, cutting false-positive threat alerts by 30% and automating security reports saving 8 hours/week of analyst effort.

Innoviti Payment Solutions Pvt. Ltd, Bangalore *Data Science & ML Engineer Intern* | Feb 2020 – May 2020 | Python · Pandas · NumPy · PostgreSQL · Jupyter

- ▶ Modelled fraud-pattern classifiers on 1M+ multi-bank payment transaction records, attaining 87% classification accuracy through feature engineering and hyperparameter tuning.
- ▶ Delivered 4 BI dashboards for transaction pattern analysis, adopted across 3 business units to optimize payment routing decisions and reduce manual reporting by 6 hours/week.

SNSE, New Delhi *Python Developer* | July 2017 – Aug 2019 | Team: 4 | Python · Django · ML · OpenCV · NLP · ROS · AWS

- ▶ Shipped 4 production systems over 2 years: NLP powered Knowledge Management Portal, AI-based rescue-robot victim detection (OpenCV + ROS), collaborative filtering recommendation engine (92% precision), and ETL microservices serving 5+ enterprise BI clients.
- ▶ Optimized face-recognition and object-localization module for autonomous rescue robots in low-light and occluded environments, reducing missed-identification rate by 20% and reaching real-time inference at 15 FPS with ROS integration.

EDUCATION

M.Tech in Information Technology IIIT Allahabad (Indian Institute of Information Technology), Uttar Pradesh | 2019

Research: DNA & Genome Sequencing, Olfaction Detection via ML, Pattern Matching

Professional Certificate Program in Business Management (PCPBM) IIM Kozhikode (Indian Institute of Management), Kerala | 2022

CERTIFICATIONS

- ▶ **Generative AI Leader** Google Cloud | 2026.
- ▶ **Generative AI for Everyone** Coursera | 2024.
- ▶ **Machine Learning Specialization** Coursera (Andrew Ng) | 2023.
- ▶ **Chatbot Builder Training** Yellow.ai Academy | 2022.
- ▶ **Data Science Web App Streamlit** Coursera | 2020.
- ▶ **Python for Data Science** Intellipaat | 2020.
- ▶ **Advanced Java Certification** C-DAC Jaipur | 2015.

ACHIEVEMENTS & PUBLICATIONS

- ▶ IEEE Publication: "Predicting Smell Perception from Molecular Descriptors using Machine Learning," IEEE En&T 2020, Dolgoprudny, Russia. ISBN: 978-1-7281-8830-0, Page 331.
- ▶ Paper: "Mobile Virtual Class," Imperial Journal of Interdisciplinary Research (IJIR), 2017, Vol-3, Issue-4.
- ▶ Best Paper Award: "iDoc," National Conference on Smart Computation and Technology (NCSCT), Poornima Institute, Jaipur, April 2016. Published in IJCTT V35(5).
- ▶ Organizer: International Conference on Bioinformatics and System Biology 2018, IIIT Allahabad.